

MSAS – Group Work – 18

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1 Exercises

Take the first letter of your surnames and convert them to a number following the alphabetical order (e.g., A = 1, B = 2, ..., Z = 26). Call these parameters a , b , and c . As an example, the group work of Bianchi, Neri, and Rossi leads to $a = B = 2$, $b = N = 14$, and $c = R = 18$. These parameters will be used in the following exercises.

Exercise 1

The electro-mechanical system in Figure 1 is constituted by an electric circuit that drives a mechanical part. A voltage source $v(t) = v_0 \cos(\omega t) e^{-\beta t}$ is activated at $t = 0$. Assume a coefficient K_m such that $T(t) = K_m i(t)$ and $e(t) = K_m \Omega_1$, where T is the torque acting on the mechanical part, i the current in the circuit, and Ω_1 the angular velocity of the first disk. Considering $K_m = 100/a$; $R = 200 + b + c$; $v_0 = 2$ V; $\omega = 5$ Hz; $\beta = 0.2$ Hz; $L = 2$ mH; $J_1 = 0.5$ kg m²; $J_2 = 0.3$ kg m²; $b = 0.1$ kg m² s⁻¹; $k = 0.5$ Nm, it is asked to:

- 1) Derive the mathematical model of the system;
- 2) Determine the system eigenvalues;
- 3) Select and motivate the most appropriate integration scheme;
- 4) Show the system response until $t = 30$ s;
- 5) Setup, discuss, and run a procedure to find the values of K_m and R that allow matching the second disk angular velocity profile sampled at 10 Hz given in the file `Profile.txt`

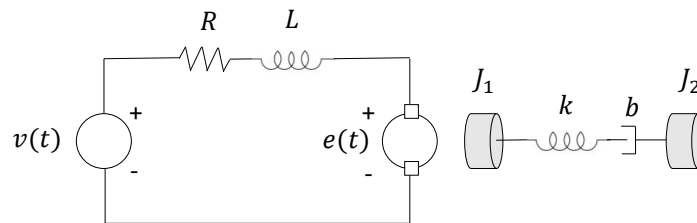


Figure 1: Electro-mechanical system.

1.1 Physical model

The first step of the dynamical investigation implies deriving a physical model for the real system under study.

The actual system can be thought as a permanent magnets DC motor, which to be studied requires coupling two different engineering fields: in this case it involves the mechanical and electrical domains.

In order to simplify the approach we used a lumped parameters model for the armature circuit and a rigid body behaviour for the mechanical side of the system.

Moreover, assuming both elastic and damping effects to be linear, Figure 2 shows the resulting physical model.

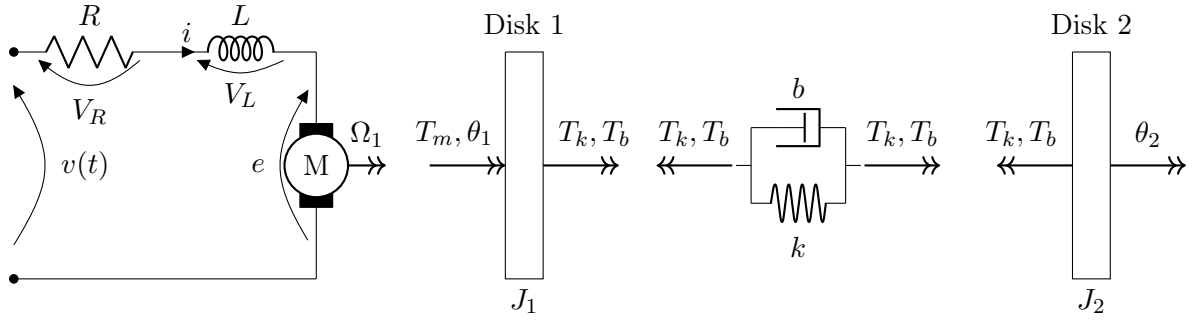


Figure 2: Physical model of each system component

1.2 Mathematical model

Still referring to Figure 2, we can derive the mathematical model using two main tools: Kirchhoff's voltage law for the electrical circuit and the torque law for each of the two disks.

In addition to them we also need some constitutive equations expressing the behaviour of the various components inside the system.

The electric components are modelled as follows:

- Resistor: $V_R(t) = Ri(t)$
- Inductor: $V_L(t) = L \frac{di}{dt}$
- Electromotive force: $e(t) = K_m \Omega_1 = K_m \dot{\theta}_1$

For what regards the mechanical behaviour, both the spring and the damper have been assumed to be linear objects; therefore their characteristic equations are:

- Spring: $T_k(t) = k(\theta_2 - \theta_1)$
- Damper: $T_b(t) = b(\dot{\theta}_1 - \dot{\theta}_2)$

having taken, without loss of generality, $\theta_2 > \theta_1$ and $\dot{\theta}_2 > \dot{\theta}_1$.

Then obtaining the governing equations for the system dynamics is quite straightforward and the result is condensed in Equation 1.

$$\begin{cases} L \frac{di}{dt} + Ri + K_m \Omega_1 = v(t) \\ J_1 \ddot{\theta}_1 = k(\theta_2 - \theta_1) + b(\dot{\theta}_2 - \dot{\theta}_1) + \frac{K_m}{J_1} i \\ J_2 \ddot{\theta}_2 = -k(\theta_2 - \theta_1) - b(\dot{\theta}_2 - \dot{\theta}_1) \end{cases} \quad (1)$$

The system above is composed by three linear ordinary differential equations with constant coefficients, the main difference between them being that two are second order ODEs while the other one is first order.

These peculiarities allow us to rewrite the set of equations in a state-space representation with five system states to be chosen:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{u}(t)$$

If we select the state vector as $\mathbf{x} = [i, \theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2]^T$ we can express the equations of motion in the following matricial form:

$$\begin{Bmatrix} \frac{di}{dt} \\ \dot{\theta}_1 \\ \ddot{\theta}_1 \\ \dot{\theta}_2 \\ \ddot{\theta}_2 \end{Bmatrix} = \begin{bmatrix} -\frac{R}{L} & 0 & -\frac{K_m}{L} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ \frac{K_m}{J_1} & -\frac{k}{J_1} & -\frac{b}{J_1} & \frac{k}{J_1} & \frac{b}{J_1} \\ 0 & 0 & 0 & 0 & 1 \\ 0 & \frac{k}{J_2} & \frac{b}{J_2} & -\frac{k}{J_2} & -\frac{b}{J_2} \end{bmatrix} \begin{Bmatrix} i(t) \\ \theta_1 \\ \dot{\theta}_1 \\ \theta_2 \\ \dot{\theta}_1 \end{Bmatrix} + \begin{Bmatrix} \frac{v(t)}{L} \\ 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix} \quad (2)$$

1.3 Selecting an integration scheme

In order to chose a proper numeric integrator we should first study the stability nature of our system.

Since by definition of stability one does not have to account for the forcing term, we only need to check the eigenvalues of the state matrix $\mathbf{A}$ expressed in Equation 2.

Figure 3 shows their location in the complex plane. It is evident that the system can be classified as stiff, being one eigenvalue smaller than the other four by several orders of magnitude.

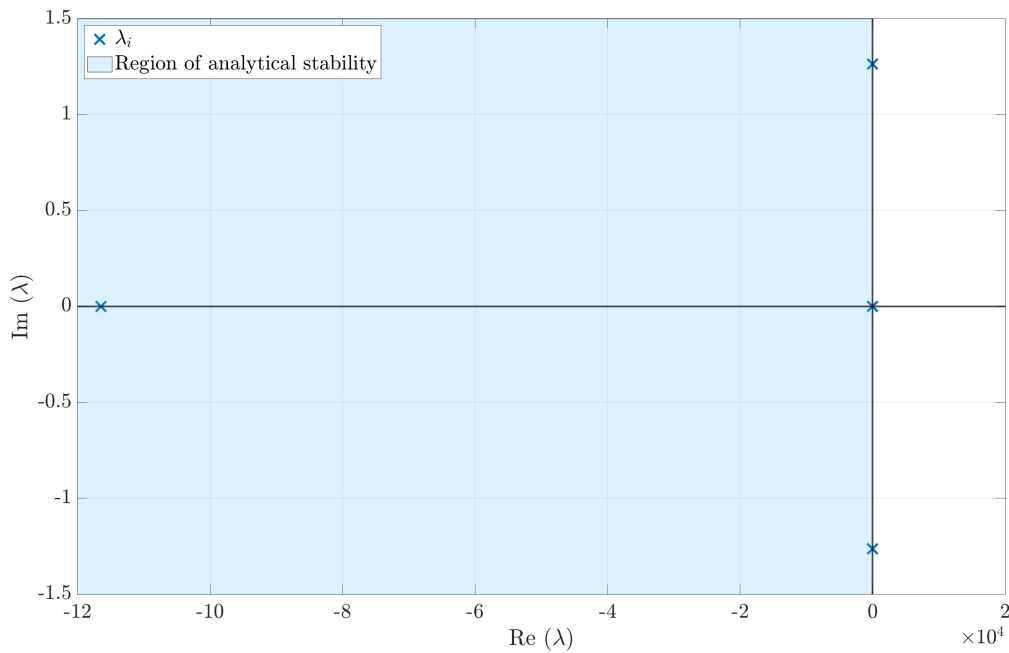


Figure 3: State matrix eigenvalues

To better understand the spectral distribution of the state matrix $\mathbf{A}$, Table 1 lists all the numerical results of the eigenvalue computations.

We can notice that, except for the first one, they are all quite close to the origin; moreover two of them are complex conjugates therefore the free response of the states will exhibit an oscillating behaviour.

Being λ_1, λ_2 and $\lambda_{4,5}$ characterized by a negative real part, they fall into the region of analytical stability.

More of a concern is the resulting value for λ_3 : even though its magnitude makes it negligible and we might think of it as being different from zero only for numerical reasons, we tried and

investigate if it would pose a threat on the system stability.

Table 1: State matrix eigenvalues

	Value
λ_1	$-1.165 \cdot 10^5$
λ_2	-5.379
λ_3	$3.4 \cdot 10^{-18}$
$\lambda_{4,5}$	$-0.259 \pm 1.263i$

As a consequence we selected an implicit A-stable method, in which theoretically positive eigenvalues cause instability, and we simulated the free response of the system with non-homogeneous initial conditions.

Figure 4 shows that in the required time span no significant divergent behaviour emerges: the states either converge to zero or to a different but still constant value. Thus we can enforce $\lambda_3 = 0$ without issues.

Eventually it means that the only requirement we have on the integrator is to choose an A-stable one such that the integration step h doesn't become excessively small because of the presence of λ_1 .

Moreover we can observe that the electric current has the fastest dynamics inside the system: we can associate it to the smallest eigenvalue that makes the system stiff; in the control system domain it would be said that $[\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2]$ are the dominant poles.

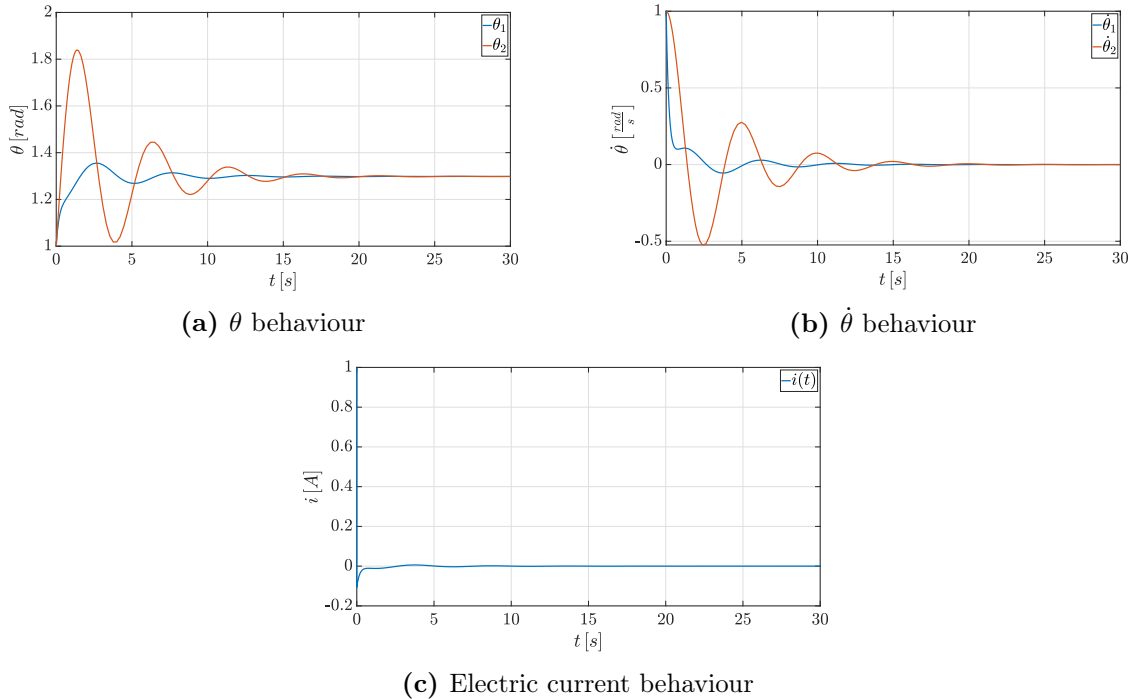


Figure 4: Free response of the electro-mechanical system

1.4 System response

Given the mathematical model derived in Section 1.2, our constants $K_m = 25, R = 233$ and having selected an implicit backinterpolation technique, Figure 5 depicts the response of the system under the assigned input voltage.

We can observe that due to the input behaviour in time, i.e. an harmonic function whose amplitude exponentially decreases over time, all the states eventually reach a steady state. This is something peculiar: usually for rotational system even if the angular velocities settle on a constant value the angular coordinates would continue increasing over time. The reason for this not to occur is that first of all the rotational speeds both tend to zero, meaning that the angles won't change anymore once the steady state condition has been reached; moreover the oscillating behaviour of the voltage makes all the states change the sign thus the values of $\theta_{1,2}$ will be limited in amplitude.

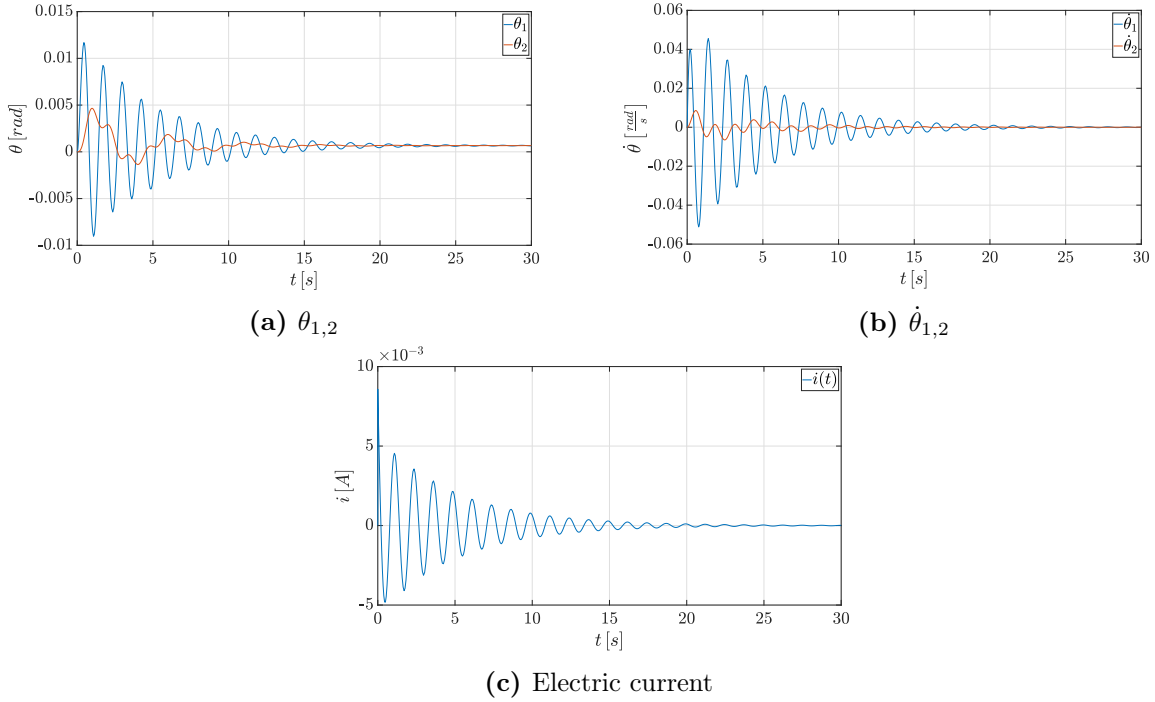


Figure 5: Simulated response to the forcing action of $v(t)$

1.5 Matching experimental data

The next problem to be solved requires finding and selecting proper values for K_m (and) R to properly catch the experimental response of $\dot{\theta}_1$ to the same input $v(t)$.

The starting point is to define a proper *merit index*, i.e. a function, depending on both the experimental samples and the constants under investigation, to be minimized.

Since in the numerical domain we work with vectors, i.e sets of values, our idea was to select a parameter that inherently accounts for all the components and not for just one.

The main reason for such choice being that since experimental samples might be affected by random noises and disturbances, focusing on the whole set of data instead of one alone, would deliver a much better global approximation.

That said, our explicit definition of the merit index is:

$$error = \|\dot{\theta}_1^{experimental} - \dot{\theta}_1^{simulated}(K_m, R)\|_2 = \sqrt{\sum_{i=1}^N \left(\dot{\theta}_{1,i}^{experimental} - \dot{\theta}_{1,i}^{simulated} \right)^2} \quad (3)$$

In the following sections we will describe the three methods we have used to minimize the *error*, each one working with a different base mechanism.

1.5.1 Grid Search

Having defined the error in Equation 3, we first wanted to have a look at its behaviour in the (K_m, R) space to get a better understanding of the region of interest.

The so-called Grid Search approach splits each independent variable domain into a finite vector having a certain number of elements; in this case we will have two vectors representing K_m, R having respective lengths N_k, N_R . Then for each component along one dimension it cycles through all values of the other one.

Such an approach produces a $N_k \times N_R$ matrix which can then be used to visualize the function, as shown in Figure 6.

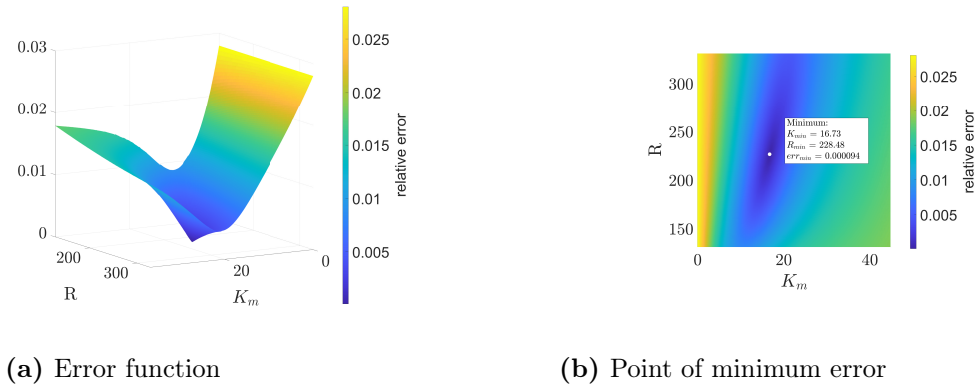


Figure 6: Graphical results obtained through the Grid-search approach

Despite the advantage of delivering a graphical results such an approach is extremely computationally intensive: in order to get a well refined result over wide domains the number of points used skyrockets.

The plots shown in the previous Figure were created with a $[200 \times 200]$ matrix; indeed it's a noticeable burden on the computing hardware.

However in this specific case it can be noticed that each cycle is completely independent from the other ones: therefore there is the possibility of exploiting the multi-core composition of moderns CPUs to run different loops in parallel.

This leads to a spread of the computational weight that eventually results in a much lower execution time.

1.5.2 Gradient-based approach

A different way to minimize a generic function relies on using the Matlab function `fmincon`. Default tolerances are selected and only a lower bound is defined, i.e. $[0, 0]$, as having negative K_m or R would not make physical sense. This method is the most efficient of the three used in this analysis, the only issue with it being that the result is very sensitive to the initial conditions provided. Despite that, defining $K_m = 25$ and $R = 233$ as initial conditions, the error associated to this solution is smaller than that of the other methods of this analysis, so the minimum found is most likely the global one.

1.5.3 Genetic algorithm

Genetic algorithms work in a totally different way: they generate a random initial set of points in the domain, called *population* and for each of them a *fitness value* is computed.

Such value is determined by several parameters, among them being the local value of the function to be minimized. After this step only a subset of the initial population is retrieved, containing the points which exhibit the best fitness, and a new random population is created

around these surviving elements.

The process then continues until either a maximum number of iterations is reached or the tolerance falls under a given value.

These algorithms are very effective in determining the minimums of the function under investigation when dealing with extended areas; however due to their intrinsic random nature they might easily get caught into local minimums without having the ability to escape.

For this reason it is good practice to make a genetic algorithm run several times before retrieving a final value; nonetheless they can also be used to find a suitable starting point for more refined techniques such as the one described in Section 1.5.2.

1.5.4 Summary and final simulation

Table 2 shows the results achieved with the mentioned minimization techniques, along with the condition given by the initial values. Interestingly we were already quite close to the experimental samples with the provided values of (K_m, R) .

Let's make a further brief consideration on the applied techniques: the Grid search and the Inverse gradient methods provide accurate and repeatable results still being the second one usually much faster, instead the reported error for Genetic algorithm considered the best case scenario among all simulations performed. Eventually the orders of magnitude for the errors

Table 2: Minimization results

	K_m	R	error
Initial values	25	233	$8.70 \cdot 10^{-3}$
Grid search	16.73	228.47	$9.41 \cdot 10^{-5}$
fmincon	16.70	227.75	$8.30 \cdot 10^{-5}$
Genetic algorithm	16.72	228.12	$8.62 \cdot 10^{-5}$

were deemed more than acceptable: since experimental samples are often affected by random disturbances one can never achieve a perfect matching.

Moreover we should also account for the limits of our model: to better catch the real world dynamics considering deformable bodies instead of rigid ones might improve the accuracy even though in this case the additional complexity it wouldn't be worth the cost.

Figure 7 and 8 shows our final results, with values for the constants taken as $K_m = 16.70$, $R = 227.75$.

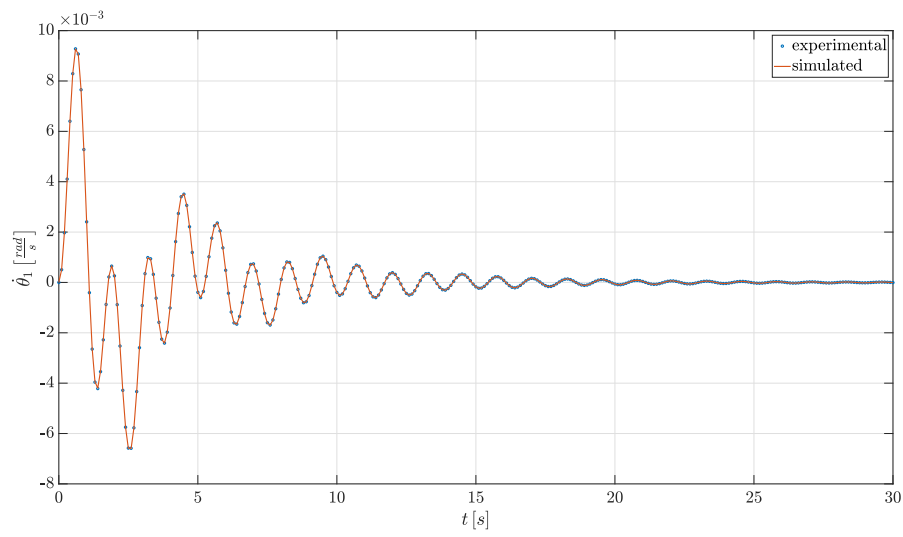


Figure 7: Experimental and simulated results

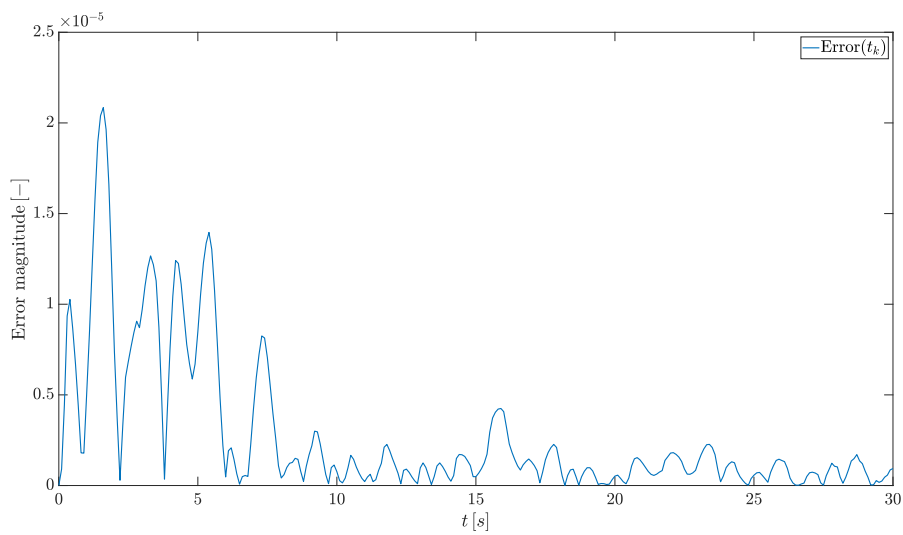


Figure 8: Absolute value of the error at each sampling time

Exercise 2

The hydraulic system in Figure 9b is made of a tank, a pump, a check valve, a distribution valve, a filter and two heat exchangers, plus the lines. The heat exchanger on the left is used to cool down an external system, having a temperature profile $T(t) = T_0 + k_T \cos(\omega t)$, and its wall is made up of three layers with different thermal properties, as depicted in Figure 9a. The exchanger on the right (cooler) removes the excessive heat from the fluid with a constant heat flux $\dot{Q}_c$. The pressure drop inside the heat exchangers can be modelled as a simplified Rayleigh flow, such that $P_{out} = e^{\dot{Q}/\kappa} P_{in}$. Assuming:

- Fluid: Water, incompressible fluid, $\rho = 1000 \text{ kg/m}^3$, specific heat $c_w = 4186 \text{ J/(kg} \cdot \text{K)}$
- Lines: Coefficient of pressure drop across the check valve $k_{cv} = 2$, diameter of the lines $D = 20 \text{ mm}$;
 - Branch T–1: Length $L_{T1} = 0.5 \text{ m}$, friction factor $f_{T1} = 0.032$;
 - Branch 3–4: Length $L_{34} = 1.5 \text{ m}$, friction factor $f_{34} = 0.032$;
 - Branch 5–6: Length $L_{56} = 0.2 \text{ m}$, friction factor $f_{56} = 0.040$;
 - Branch 7–8: Length $L_{78} = 2.5 \text{ m}$, friction factor $f_{78} = 0.035$;
 - Branch 9–10: Length $L_{91} = 2.5 \text{ m}$, friction factor $f_{91} = 0.028$;
 - Branch 11–T: Length $L_{11} = 1 \text{ m}$, friction factor $f_{11} = 0.032$.
- Tank: Adiabatic tank with constant pressure $P_T = 0.1 \text{ MPa}$;
- Pump:

Pistons:

- Number: $N = 9$;
- Diameter: $D_p = 1.5 \text{ cm}$
- Diameter of the shaft: $d_p = 0.7 \text{ cm}$
- Nominal pressure: 5 atm

Pilot piston:

- Diameter: $D_k = 1 \text{ cm}$
- Control lever length: $l_c = 10 \text{ cm}$
- Maximum angle of the control plate: $\theta_{\max} = 20 \text{ deg}$
- Rotation speed: $n = 4000 \text{ rpm}$
- Equivalent mass: $m_k = 2 \text{ kg}$
- Pre-loaded force: $F_0 = 5 \text{ N}$
- Friction coefficient: $r_k = 1 \text{ Ns/m}$
- Diameter of the pipe: $d_k = 1 \text{ mm}$
- Pilot pipe head loss: $k_p = 2.5$
- Distributor: Coefficient of pressure drop across the distributor $k_d = 15$, diameter $d_o = 10 \text{ mm}$. At $t_0 = 0 \text{ s}$ the valve is half open; it is fully open after $\Delta t = 2 \text{ s}$.
- Cooler: Diameter of the pipe $D = 20 \text{ mm}$; Heat flux $\dot{Q}_c = 100 \text{ W}$.
- Filter: Coefficient of pressure drop across the filter $k_f = 35$, leaking coefficient $k_l = 2.5\%^1$;

¹ $Q_{leak} = k_l Q_{in}$

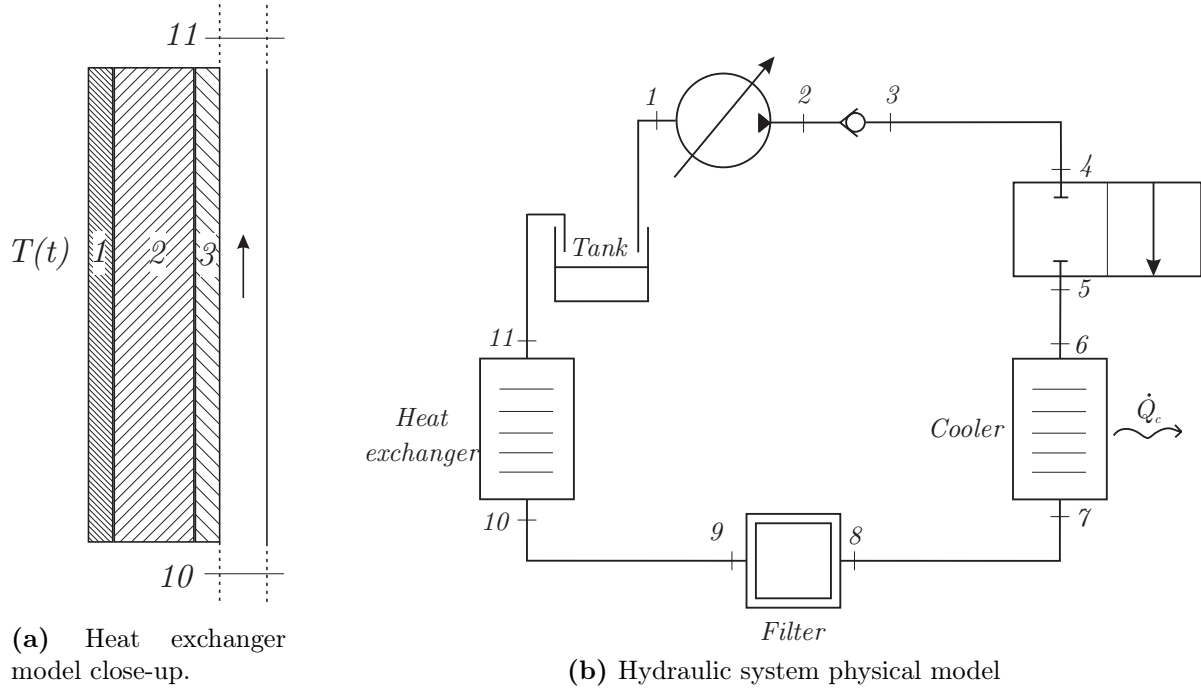


Figure 9: Thermo-hydraulic system. Assume any other missing data.

- Heat exchanger: Diameter of the pipe $D = 20$ mm, length of the pipe inside the exchanger: $L_e = 0.5$ m; exchange area $A_e = 1000$ cm²; First layer thermal properties: $k_1 = 395$ W/(m · K), $\ell_1 = 1$ cm, second layer thermal properties: $k_2 = 310$ W/(m · K), $\rho_2 = 8620$ kg/m³, $c_2 = 10a$ J/(kg · K), $\ell_2 = 2.5$ cm, third layer thermal properties: $k_3 = 125$ W/(m · K), $\ell_3 = 1$ cm, heat transfer coefficient with the fluid $h = 20$ W/(m² · K);
- $T_0 = 400$ K, $k_T = 2b$, $\omega = 5$ s⁻¹, $\kappa = 100c$ W;
- The temperatures are propagated instantaneously along the pipes;
- The heat exchanger layers have an initial temperature of 340 K.

It is asked to:

- 1) Derive a lumped-approach physical model for the heat exchanger;
- 2) Derive the mathematical model of the whole system;
- 3) Select and motivate the most appropriate integration scheme;
- 4) Show the system response until $t = 25$ s;
- 5) Discuss at least one possible way to modify the system in order to keep the fluid temperature as close as possible to 20 °C.

2.1 Physical model

To create a physical model of the system under investigation, we consider as the main driver the hypothesis for which the pipes allow the instantaneous propagation of temperatures along the system. As a first consequence, we can immediately say that the reference temperature of our system will be only one. It is true, in fact, that in every instant of time, we would have a higher temperature and a lower temperature across the heat exchangers. However, these two

temperatures allow us to define a ΔT temperature difference which will be the same for both the heater and the cooler. Obviously, we know that this difference will vary over time because it depends on the flow rate that flows through the system. Moreover, the latter will also depend on the behavior of the pump.

The pump is equipped with a pilot piston which behaves like a second order mechanical system: there is, therefore, an inertia which involves, from the point of view of the mathematical model, a non-algebraic evolution of the flow rate as a function of the other parameters that characterize the problem. In this case, the main driver of the flow rate variation in the system is the pressure that governs the motion of the pump pilot piston.

Furthermore, in the description of the model we observe that there is no heat exchange with the external surroundings, with the exception of the heat exchangers. This means that thermal power is conserved: more precisely, we are saying that the thermal power entering the system is either absorbed by the fluid or dissipated by other components of the system. In fact, there is a loss of flow rate in the filter due to leakage which results in a loss of thermal energy.

At this point, knowing the volumetric flow rate of fluid, we can determine the ΔT temperature difference. From the latter it is possible to obtain the thermal power which, passing through the exchanger, enters directly into the fluid.

Given this physical analysis of the problem, we can build a complete model for the heat exchangers where all the boundary conditions are assigned. This model is fully illustrated in Figure 10. The heat exchanger has a cylindrical geometry, and it is characterized by two

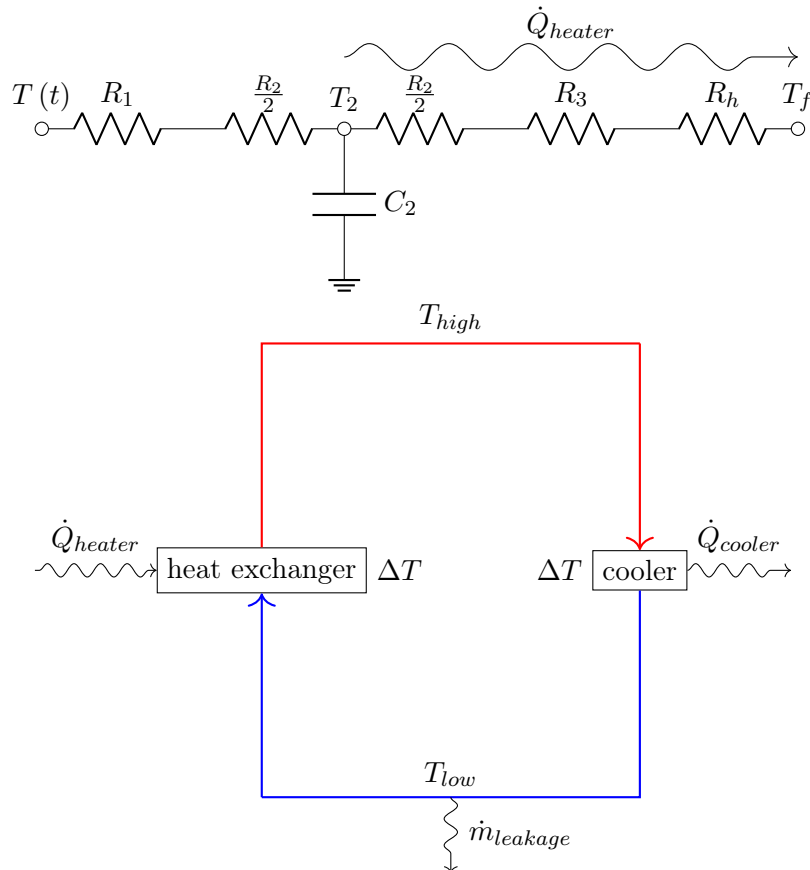


Figure 10: Physical model for the heat exchanger and the thermal circuit

completely thermal resistive layers and by an additional layer characterized also by a thermal capacity. Furthermore, the thermal source has an oscillatory trend over time. Inside the heat exchanger, only one temperature is required as a state variable, as all other temperatures are obtained from algebraic relations.

The conductive nodes are positioned in the middle of each layer while the node associated with the temperature of the fluid is positioned in such a way to require the entire convective resistance to be crossed.

Finally, the temperature of the fluid is calculated by imposing that the heat flowing between the node T_2 and the node T_f is the one resulting from the conservation of the temperature difference ΔT across the heat exchangers.

From the point of view of the fluid dynamics system, we can add that the pressures that characterize the different sections of the system are calculated independently of the pump. The pump outlet pressure, in fact, is that one necessary to get the fluid to the tank. Consequently, we calculate the pump outlet pressure by going up the circuit from the tank, taking into account the distributed and concentrated head losses. In detail, the pump is characterized by an angular velocity which involves how much volume of fluid is moved by the pump every second (volumetric flow rate), but the pressure is calculated considering the dynamic equilibrium of the system. In summary, we are determining the pressure that allows the system not to violate the conservation of mass. The latter, since the density is constant, involves the conservation of the volumetric flow rate.

Finally, the spring stiffness of the control piston has been calculated by making sure that the pump stops delivering fluid flow when the nominal pressure of 5 atm is reached. However, the pump never stops working because the pressure of 5 atm is never reached. Obviously, the higher the pressure, the lower the delivery flow rate from the pump.

2.2 Mathematical model

The state variables of the system are 4. The first two are the position and velocity of the pilot piston: these quantities are related to the mechanical evolution of the system and are regulated by a differential equation in which there is an inertia. The third state variable is the temperature T_2 which is also equipped with an inertial term represented by the thermal capacity. The last variable is the volume of the tank: it is not necessary to solve the equations of the system, but we take it into consideration in order to know how the total volume of the system varies since there are leaks.

The differential equations associated with the first two state variables described above are, now, illustrated. It should be taken into account that the derivative of the velocity of the pilot piston is obtained by taking into account the balance of forces on the system.

$$\begin{cases} \dot{x}_k = v_k \\ \dot{v}_k = \frac{1}{m_k}(P_k A_k - F_0 + h_k x_k - r_k v_k) \end{cases} \quad (4)$$

First of all, we start our analysis from the pump. We can calculate the area of the pump cylinders A_p , the area of the pilot piston A_k , and the area of the pipe which links the pump to the pilot piston $A_{k,pipe}$:

$$\begin{cases} A_p = \pi \frac{D_p^2}{4} \\ A_k = \pi \frac{D_k^2}{4} \\ A_{k,pipe} = \pi \frac{d_k^2}{4} \end{cases} \quad (5)$$

At this point, from the data we can determine the maximum stroke cc_{max} and, subsequently, taking into the account the position of the piston x_k we calculate the stroke s ;

$$\begin{cases} cc_{max} = l_c \tan(\theta_{max}) \\ s = \frac{d_{shaft}}{l_c}(cc_{max} - x_k) \end{cases} \quad (6)$$

Moreover, the stiffness associated to the pilot piston is:

$$h_k = \frac{p_{nom}A_k - F_0}{cc_{max}} \quad (7)$$

Finally, the velocity of the fluid connected point k and 2 is:

$$v_{k,pipe} = \frac{v_k A_k}{A_{k,pipe}} \quad (8)$$

Now, we can calculate the flow rate Q_1 knowing the rotational speed n , the number of pistons of the pump N , the area of the pump cylinders A_p , and the stroke s :

$$Q_1 = nNA_p s \quad (9)$$

Then, due to the leakage, the flow rate Q_9 after the filter will be:

$$Q_9 = Q_1(1 - k_{leakage}) \quad (10)$$

Taking into account the physical model described in the previous section we can now determine the values of the pressures in the different sections of the system:

$$p_1 = p_{tank} - \frac{1}{2}f_{T1}\rho_{water} \frac{L_{T1}}{D_{T1}} \frac{Q_1}{A_{pipe}} \left| \frac{Q_1}{A_{pipe}} \right| \quad (11)$$

$$p_{11} = p_{tank} + \frac{1}{2}f_{11}\rho_{water} \frac{L_{11}}{D_{11}} \frac{Q_9}{A_{pipe}} \left| \frac{Q_9}{A_{pipe}} \right| \quad (12)$$

where head losses have been considered.

Then, the pressure in point 10 is obtained considering the hypothesis of Rayleigh flow in the heat exchanger as follows:

$$p_{10} = \frac{p_{11}}{e^{\frac{Q_{in}}{k}}} \quad (13)$$

In addition, we can determine:

$$p_9 = p_{10} + \frac{1}{2}f_{91}\rho_{water} \frac{L_{91}}{D_{91}} \frac{Q_9}{A_{pipe}} \left| \frac{Q_9}{A_{pipe}} \right| \quad (14)$$

$$p_8 = p_9 + \frac{1}{2}k_{filter}\rho_{water} \frac{Q_9}{A_{pipe}} \left| \frac{Q_9}{A_{pipe}} \right| \quad (15)$$

$$p_7 = p_8 + \frac{1}{2}f_{78}\rho_{water} \frac{L_{78}}{D_{78}} \frac{Q_1}{A_{pipe}} \left| \frac{Q_1}{A_{pipe}} \right| \quad (16)$$

where head losses have been considered for the line 9-10 and 7-8 and minor losses due to the filter for the pressure p_8 .

Again, the hypothesis of simplified Rayleigh flow is used to determine p_6 :

$$p_6 = \frac{p_7}{e^{\frac{-\dot{Q}_{cooler}}{k}}} \quad (17)$$

For the line 5-6 and 3-4 head losses have been considered, while for the p_5 minor losses due to the distributor are considered:

$$p_5 = p_6 + \frac{1}{2} f_{56} \rho_{water} \frac{L_{56}}{D_{56}} \frac{Q_1}{A_{pipe}} \left| \frac{Q_1}{A_{pipe}} \right| \quad (18)$$

$$p_4 = p_5 + \frac{1}{2} k_{distr} \rho_{water} \frac{Q_1}{A_{distr}} \left| \frac{Q_1}{A_{distr}} \right| \quad (19)$$

$$p_3 = p_4 + \frac{1}{2} f_{34} \rho_{water} \frac{L_{34}}{D_{34}} \frac{Q_1}{A_{pipe}} \left| \frac{Q_1}{A_{pipe}} \right| \quad (20)$$

Finally, minor losses due to the check valve and the pump are taken into account:

$$p_2 = p_3 + \frac{1}{2} k_{cv} \rho_{water} Q_1 |Q_1| \quad (21)$$

$$p_k = p_2 - \frac{1}{2} k_p \rho_{water} v_p |v_p| \quad (22)$$

where p_k is the pressure of the pilot piston.

The distributor valve cross-section is modeled as:

$$A_{dist} = \frac{d_0^2}{8} (\alpha - \sin(\alpha)) \quad (23)$$

where $\alpha = 2\cos^{-1}(1 - 2|z|)$.

Focusing our attention on the thermal domain, we have modeled the heat exchanger considering a cylindrical geometry as sketched in Figure 11. We can calculate the conductive resistances associated to the three layers of the heater and, finally, the convective resistance associated to the fluid.

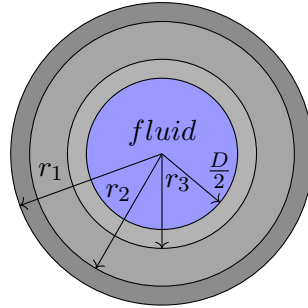


Figure 11: Heater cross-section

$$\left\{ \begin{array}{l} R_1 = \frac{\ln(\frac{D_1}{D_2})}{2\pi k_1 L_e} \\ R_2 = \frac{\ln(\frac{D_2}{D_3})}{2\pi k_2 L_e} \\ R_3 = \frac{\ln(\frac{D_3}{D})}{2\pi k_3 L_e} \\ R_h = \frac{1}{h A_e} \end{array} \right. \quad (24)$$

We can now determine the temperature difference ΔT across the cooler knowing the properties of the working fluid, the volumetric flow rate entering the exchanger Q_1 and the constant heat flux.

$$\Delta T_{cooler} = \frac{\dot{Q}_{cooler}}{c\rho Q_1} \quad (25)$$

The heat flux that the heater must provide to the colder flow is equal to that of the cooler minus the loss due to the leakage in the filter:

$$\dot{Q}_{heater} = \dot{Q}_{cooler} (1 - k_l) \quad (26)$$

As to the temperature T_2 , it has been imposed that the heat exiting from the node 2 towards the fluid must strictly depend on the exiting heat of the cooler. Since the latter is a constant, the heat flowing in the fluid will also be constant for energy conservation.

$$\begin{cases} \dot{T}_2 = \frac{1}{C_2} \left(\frac{T_{source} - T_2}{R_1 + \frac{R_2}{2}} - \dot{Q}_{heater} \right) \\ T_f = T_2 - \dot{Q}_{heater} \left(\frac{R_2}{2} + R_3 + R_h \right) \end{cases} \quad (27)$$

2.3 Integration scheme choice

This system is characterized by three different physical domains, namely the mechanical one, the fluid one and the thermal one. This would suggest that the system is stiff, but this actually is not the case, at least with the presented conditions. So the Matlab integrator `ode45` was used as it is the best compromise in terms of computational efficiency and accuracy. `ode15s` was initially tested as well, but `ode45` offered better performance overall.

2.4 System response

Below are the results of the simulation until $t = 25$ s:

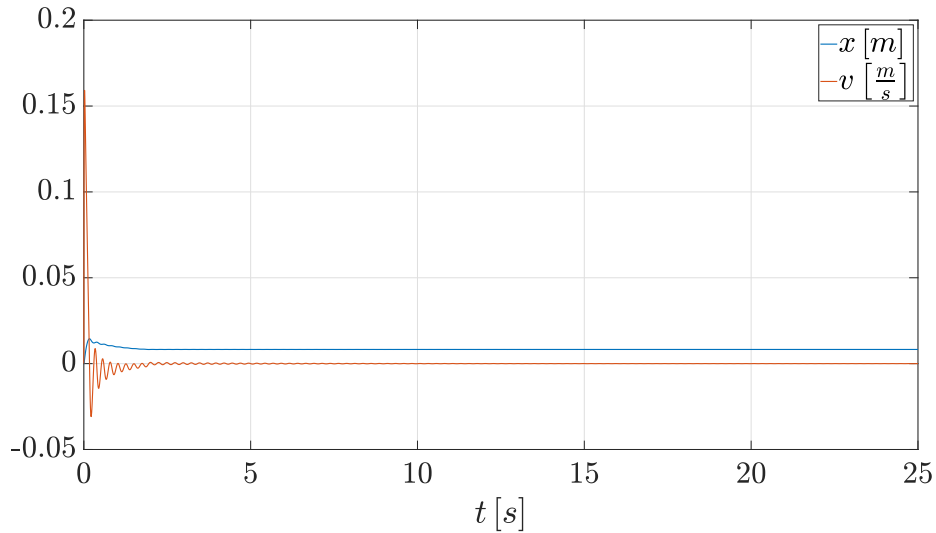


Figure 12: Position and velocity of the pilot piston head

The fluid temperature differs from the temperature T_2 by a constant. $\dot{Q}$ is constant, and the terms of thermal resistance are constant in time. We have an initial response that has a dip due to the initial conditions imposed equal to zero, while the initial acceleration of the system is not zero. The velocity, from zero, grows rapidly due to the acceleration of the system, which in turn depends on the pressure. It is interesting to note how the pressure adapts according to the flow rate that is fed into the system by the pump, which is a function of the position of the pilot piston.

It is clear from Figure 14 that the temperature variation from the hot side to the cold side is

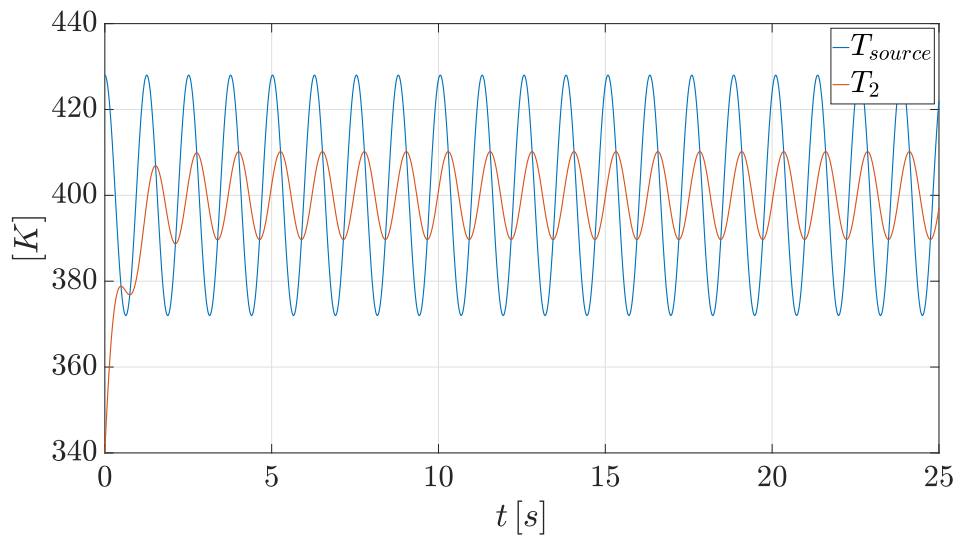


Figure 13: Temperatures inside the heat exchanger

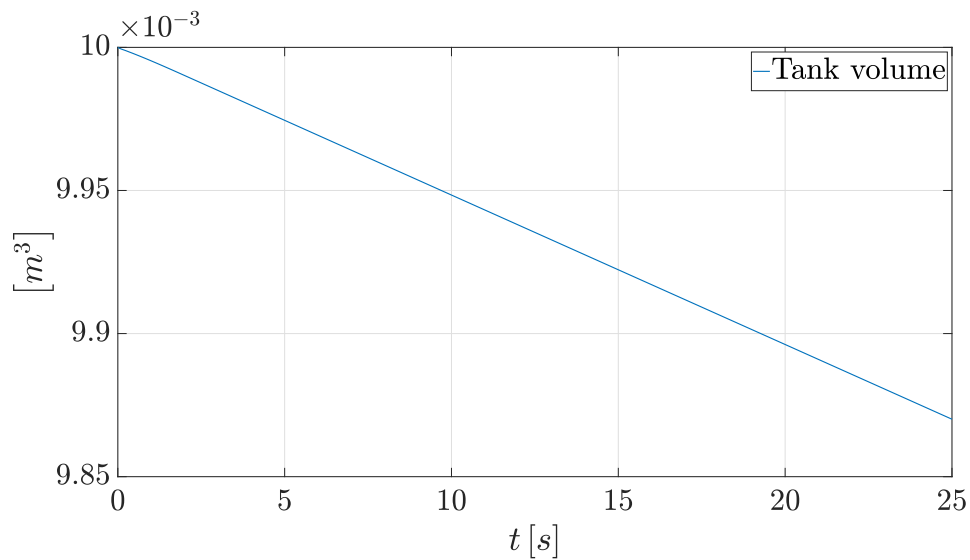


Figure 14: Volume of fluid in the tank over time

solely a function of the flow rate.

Although not very visible from the graph (Figure 16), the difference between the hot and cold temperature is a few decimals of Celsius degree. The mean temperature around which these curves oscillate is dictated by the temperature of the source and by the $\dot{Q}_{cooler}$.

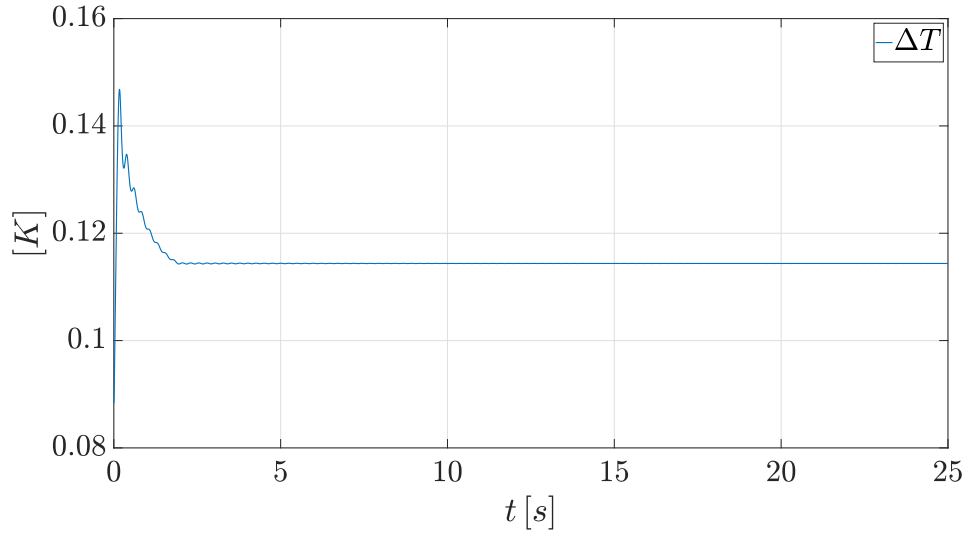


Figure 15: Fluid temperature increase with $\dot{Q}_{cooler} = 100\text{ W}$

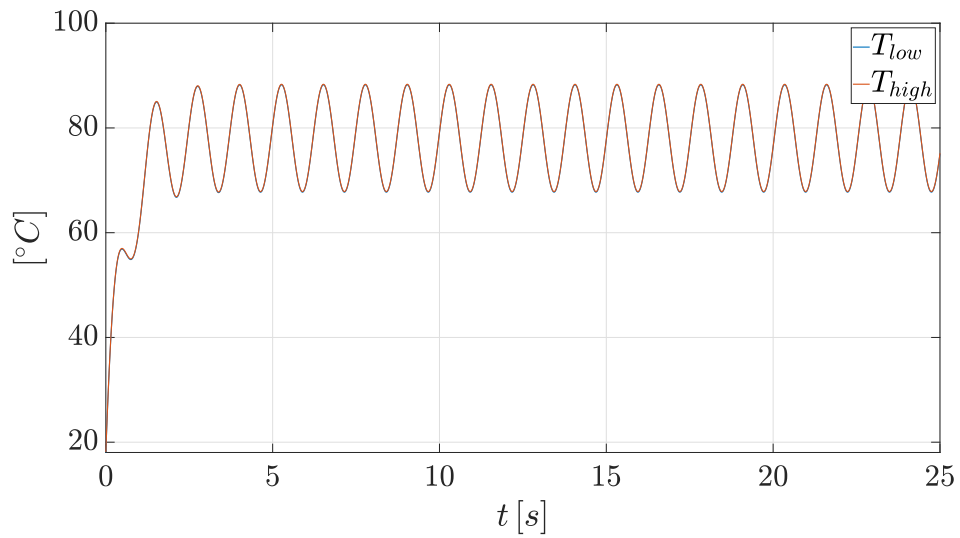


Figure 16: Fluid temperatures with $\dot{Q}_{cooler} = 100\text{ W}$

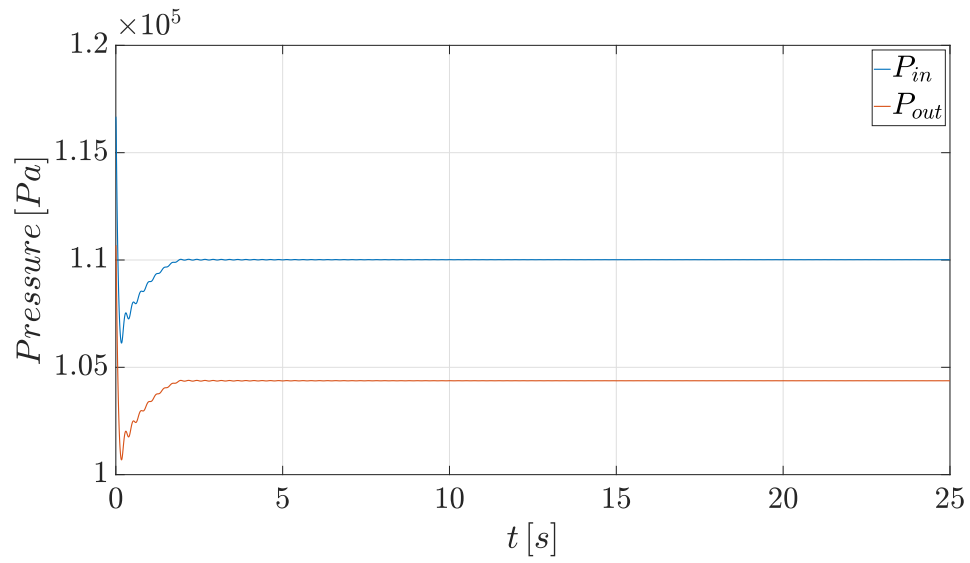


Figure 17: Pressures before and after the cooler

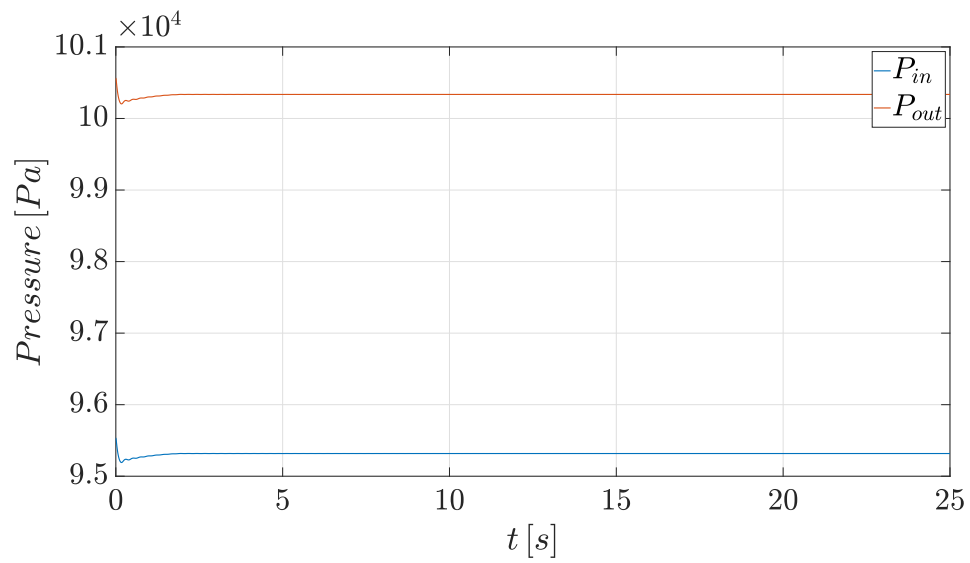


Figure 18: Pressures before and after the heater

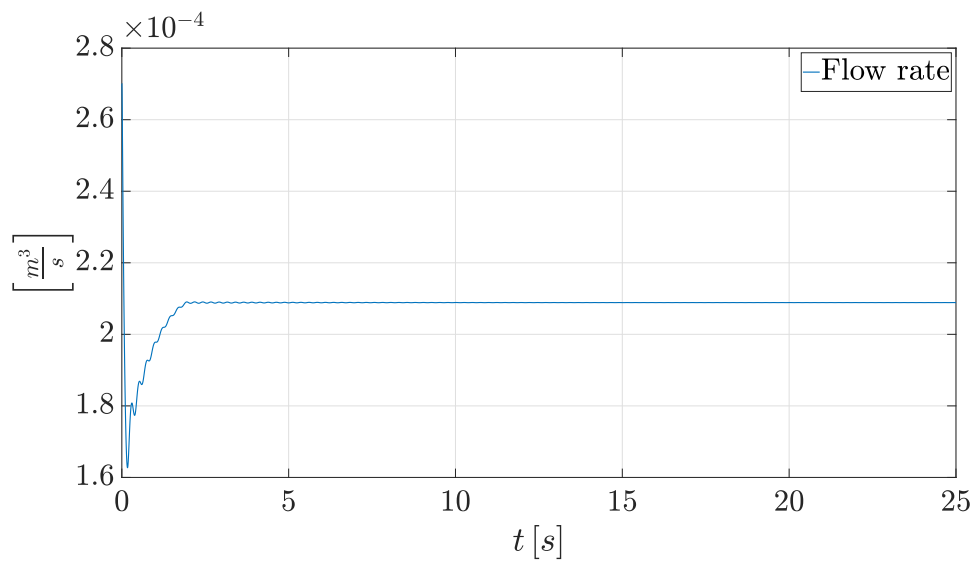


Figure 19: Flow rate over time

2.5 Temperature optimization

The analytic solution of the ODE that describes T_2 is the following:

$$T_2(t) = Ae^{-\frac{t}{B}} + \frac{k_T}{1+B^2\omega^2}\cos(\omega t) + \frac{B\omega k_T}{1+B^2\omega^2}\sin(\omega t) + T_0 + \left(R_1 + \frac{R_2}{2}\right)\dot{Q}_{in} \quad (28)$$

where:

$$\begin{cases} A = T_2(0) - T_0 + \left(R_1 + \frac{R_2}{2}\right)\dot{Q}_{in} - \frac{k_T}{1+B^2\omega^2} \\ B = C_2 \left(R_1 + \frac{R_2}{2}\right) \end{cases} \quad (29)$$

On the other hand, T_f is defined as:

$$T_f(t) = T_2(t) - \left(\frac{R_2}{2} + R_3 + R_h\right)\dot{Q}_{in} \quad (30)$$

Note that B is the time constant of the differential equation. Hence to keep T_f as close as possible to $20^\circ C = 293.15 K$ the heat $\dot{Q}_{in}$ will be:

$$\dot{Q}_{in,eq} = \frac{T_0 - 293.15}{R_1 + R_2 + R_3 + R_h} = 212.53 W \quad (31)$$

This result comes from the definition of the evolution of T_f , once the transient is over, and eliminating the oscillations forced by the temperature source. Thus $\dot{Q}_{in,eq}$ makes it so that the average of T_f after the transient will be $20^\circ C$.

But since $T_2(0) = 340 K$, while $T_0 = 400 K$, there will be a significant shift of temperature that reflects on T_f as well. The strategy adopted to combat this was to pick a time-varying $\dot{Q}_{out}$, which over time cancels out the effects of the transient of T_2 . $\dot{Q}_{out} = 100 W$ was kept as the initial value of the $\dot{Q}_{out}(t)$ profile, as that was proven to cause a $T_f(0) = 17.89^\circ C$, which is reasonably close to $20^\circ C$. Its upper bound was selected to be $\dot{Q}_{out} = \frac{\dot{Q}_{in,eq}}{1-k_l}$.

The resulting $\dot{Q}_{out}$ profile is represented in Figure 20 and written down below:

$$\dot{Q}_{out}(t) = \begin{cases} De^{-\frac{t}{B}} + E & \text{if } t \leq 5B \\ \frac{\dot{Q}_{in,eq}}{1-k_l} & \text{if } t > 5B \end{cases} \quad (32)$$

where D and E are the coefficients that verify the following:

$$\begin{cases} y(t) = De^{-\frac{t}{B}} + E \\ y(0) = 100 \\ y(3) = \frac{\dot{Q}_{in,eq}}{1-k_l} \end{cases} \quad (33)$$

In this way the transient of T_2 is practically removed from T_f so that the latter will oscillate around $20^\circ C$. The intensity of the oscillation depends on $\frac{k_T}{1+B^2\omega^2} = 3.7480$ and $\frac{B\omega k_T}{1+B^2\omega^2} = 9.5340$, as seen in Equation 28. This means that the oscillations will be of around $\pm 10^\circ C$, as shown in Figure 21. The temperature of the fluid can be kept even closer to $20^\circ C$ if these quantities are reduced, for example increasing B . This is doable by increasing the thickness of layers 1 and 2 of the heat exchanger.

In fact, if the thickness of layer 2 is increased by 2 cm, the oscillation is reduced to about $\pm 3.7^\circ C$, since $\frac{k_T}{1+B^2\omega^2} = 0.4939$ and $\frac{B\omega k_T}{1+B^2\omega^2} = 3.6859$ (see Figure 22). This choice is particularly effective because making l_2 larger affects both C_2 and R_2 . Naturally, this makes the exchanger heavier, too, in this case changing the mass of the component from 22 kg to 51.79 kg (assuming that the only part with mass is the second layer), so B cannot be increased indefinitely.

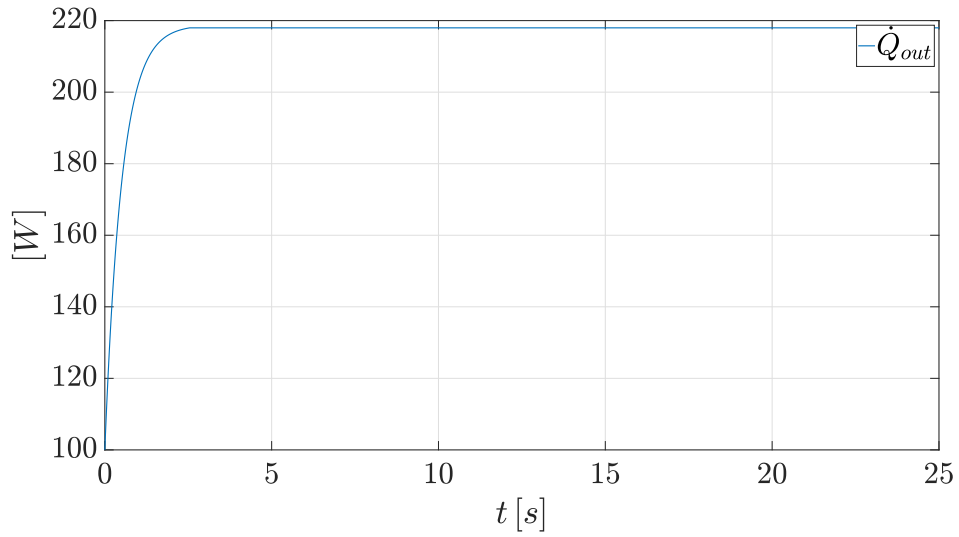


Figure 20: $\dot{Q}_{out}$ profile over time

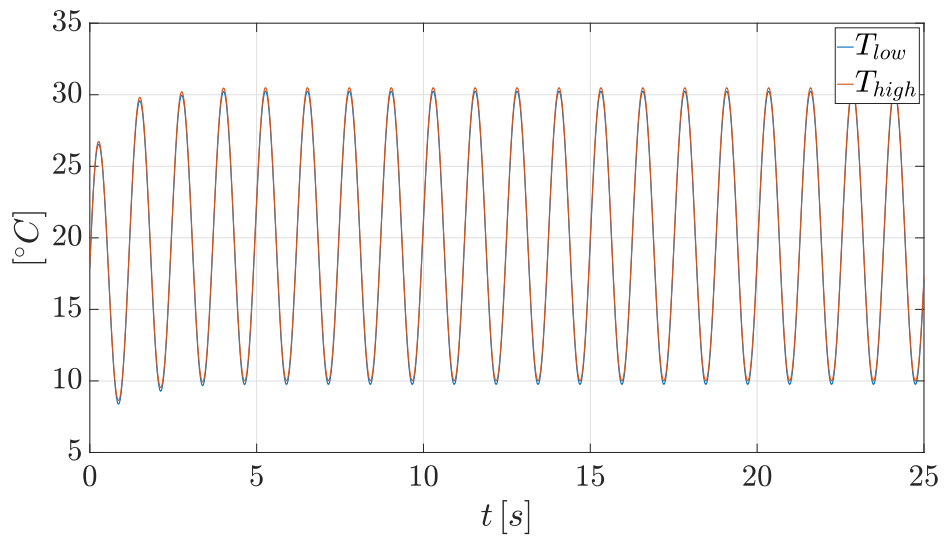


Figure 21: Fluid temperature with variable $\dot{Q}_{out}$ and $l_2 = 2.5\text{ cm}$

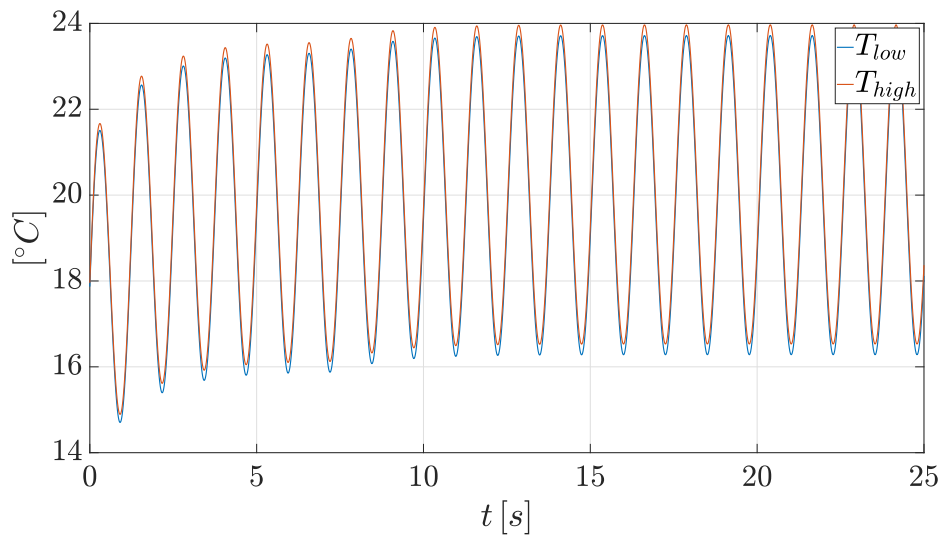


Figure 22: Fluid temperature with variable $\dot{Q}_{out}$ and $l_2 = 4.5\text{ cm}$.